

SpotterAI

AI Personal Health & Fitness Coach

SE 216 — Software Project Management

2025–2026 Spring Semester

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Project Overview

SpotterAI brings AI coaching, meal planning, workouts, and progress tracking into one app.

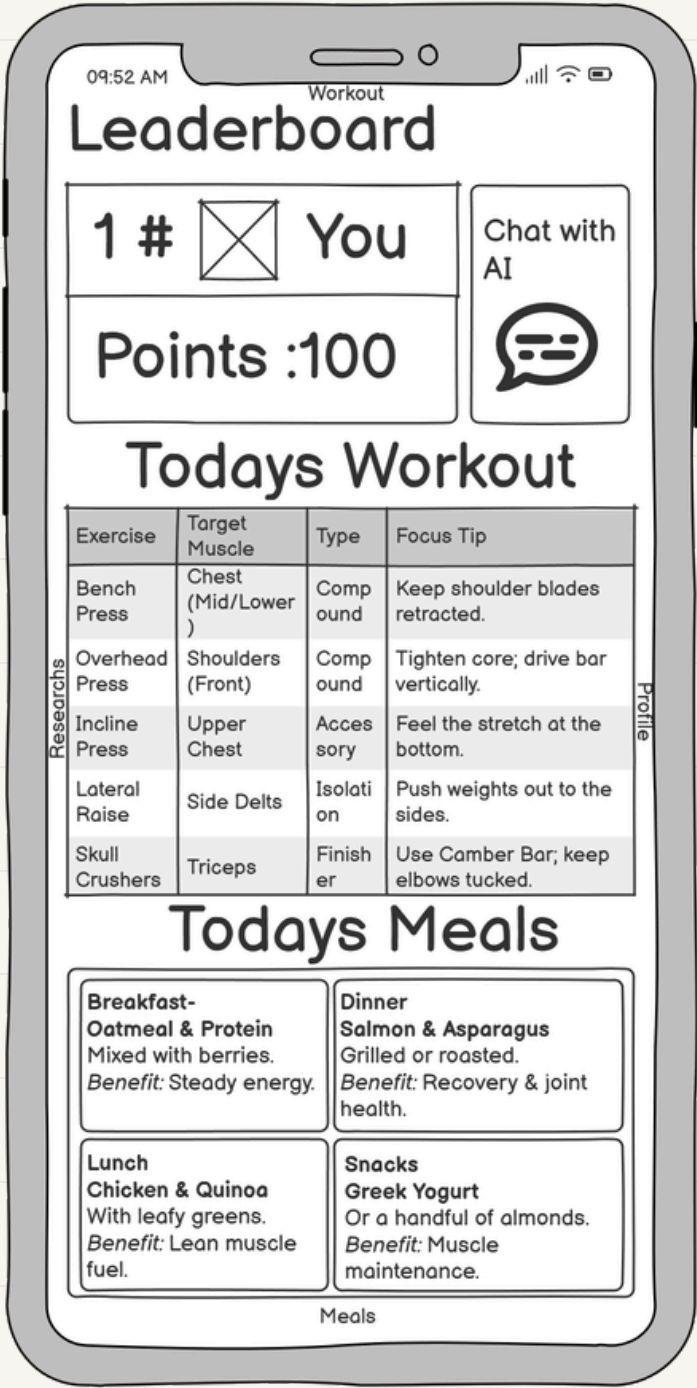
- Users often struggle to maintain consistent fitness and nutrition habits.
- Existing apps mostly give static plans and generic advice.
- SpotterAI adapts guidance through user profile data, feedback, and activity signals.

Core idea

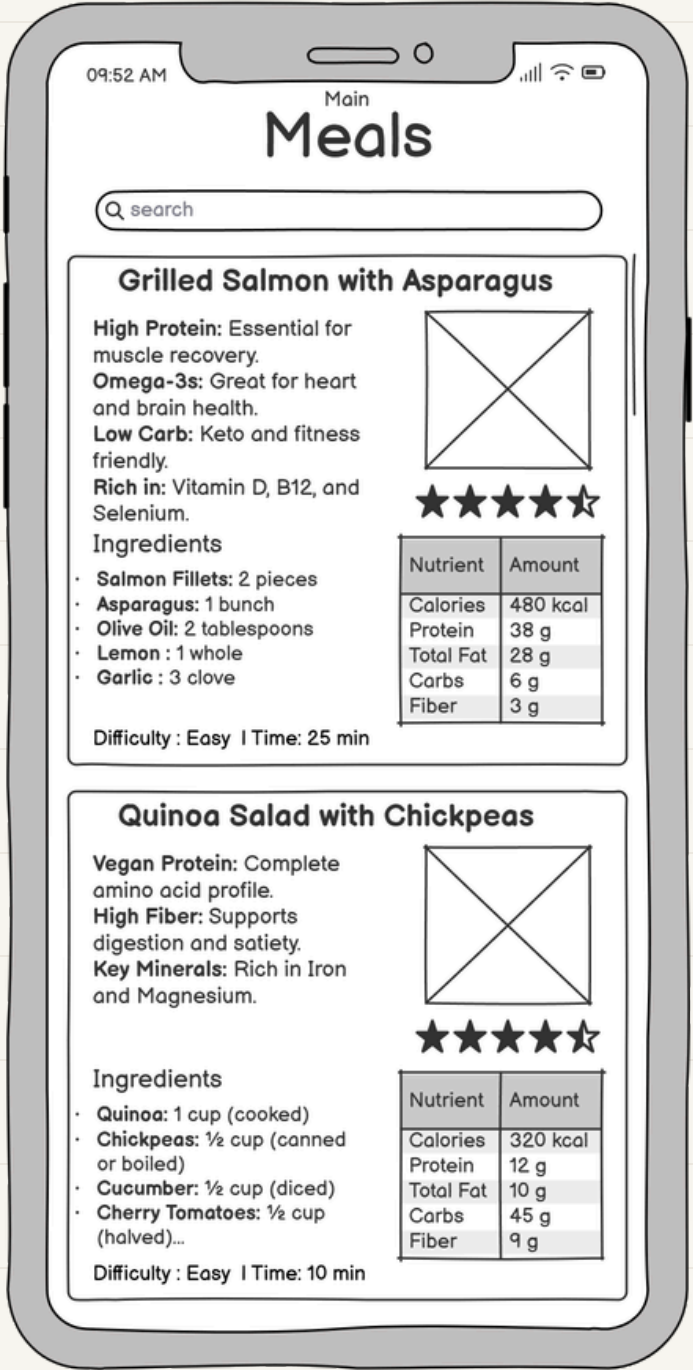
A mobile AI coach that updates daily workout and meal guidance based on user behavior.

Main value

More personal than static fitness apps and more accessible than personal coaching.



Home dashboard



Meal planning

Objectives

What the project is expected to deliver according to the report.

Personalized workout plans

Generate exercise guidance with sets, repetitions, intensity, and proper form support through Hugging Face AI.

Personalized diet plans

Create diet plans using dietary preferences, restrictions, nutritional targets, health goals, and physical characteristics.

Adaptive exercise intensity

Adjust intensity using sleep data, calorie intake, and previous performance from smartwatch and phone sensors.

Motivation and rewards

User ratings, comments, leaderboard ranking, monthly badges, and supplement discounts for top users.

Goal: provide scientifically based recommendations, monitor user behavior, and support sustainable healthy lifestyles.

Functional Requirements

Main features visible to the user.

Authentication

Users can create accounts, log in, log out, and manage their profile.

Daily AI plans

Home page shows AI-generated meal and exercise plans.

Detailed guidance

Meals include calories, macros, and allergens; workouts include sets, reps, and intensity.

Feedback loop

Ratings and comments are analyzed to improve future recommendations.

Leaderboard

Users can filter rankings by region, metric, and time period.

Rewards

Top users can receive badges or supplement discounts.

Non-Functional Requirements

Reliability, security, performance, and platform goals.

Availability

Maintain 99% uptime availability for reliable use.

Security & Privacy

OAuth 2.0, Script password hashing, AES-256 encryption, and KVKK compliance.

Platform support

iOS 13+ and Android 8+ support.

Session safety

User sessions expire after 15 minutes of inactivity.

API communication

System components communicate through RESTful APIs.

AI response time

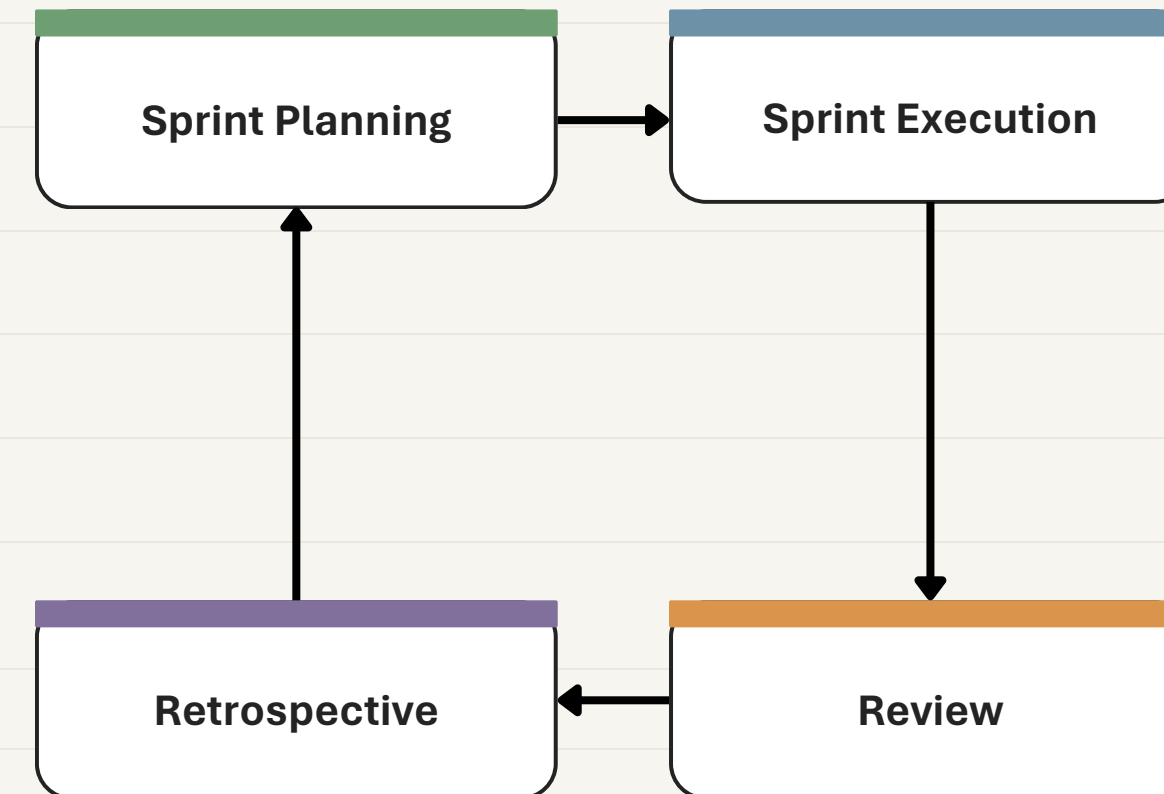
Feedback analysis should complete within 5 seconds.

Because SpotterAI handles health-related personal data, privacy and reliability are project-critical quality requirements.

Software Process Model: Scrum

A flexible model for an AI-based product.

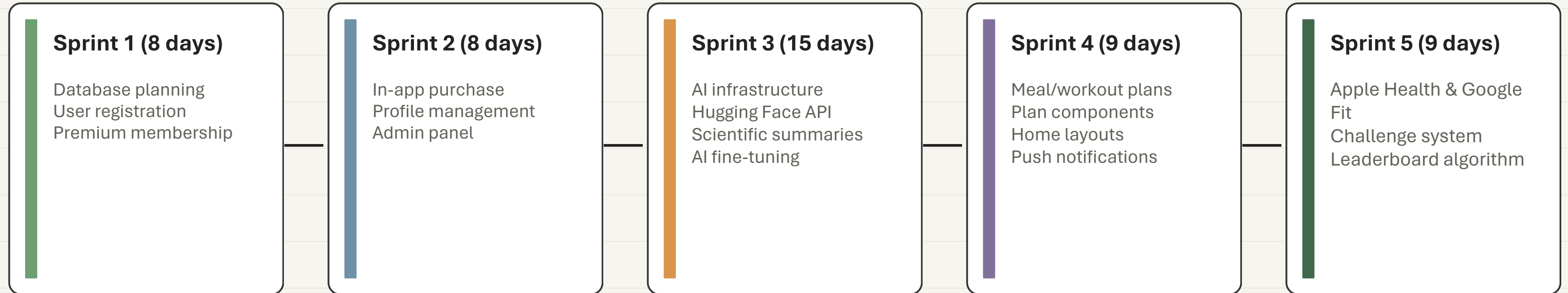
- AI integration requires continuous testing and adaptation.
- Scrum supports incremental delivery through short sprint cycles.
- Sprint reviews help the team collect feedback before the final version.
- Retrospectives help improve team communication and task planning.



Scrum lets us test assumptions early instead of discovering problems at the end.

Product Backlog & Sprint Plan

From infrastructure to intelligent user features.



The sprint plan moves from technical foundation to AI features and user motivation features.

Software Tools

Selected tools and why they fit the project.



Hugging Face

Selected for the AI recommendation engine because it supports pre-trained models, fine-tuning, and low initial cost.



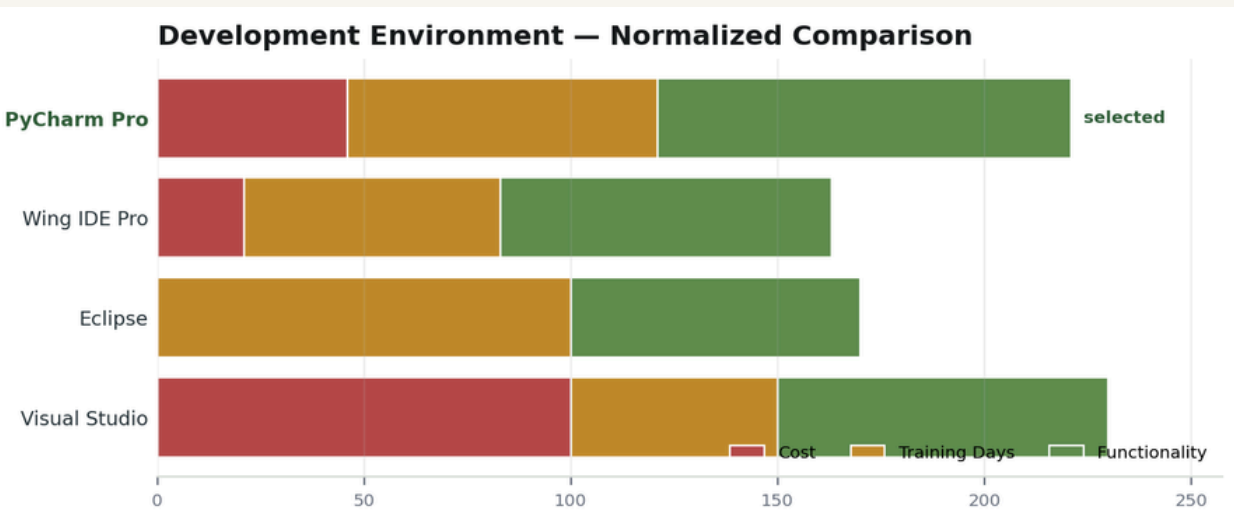
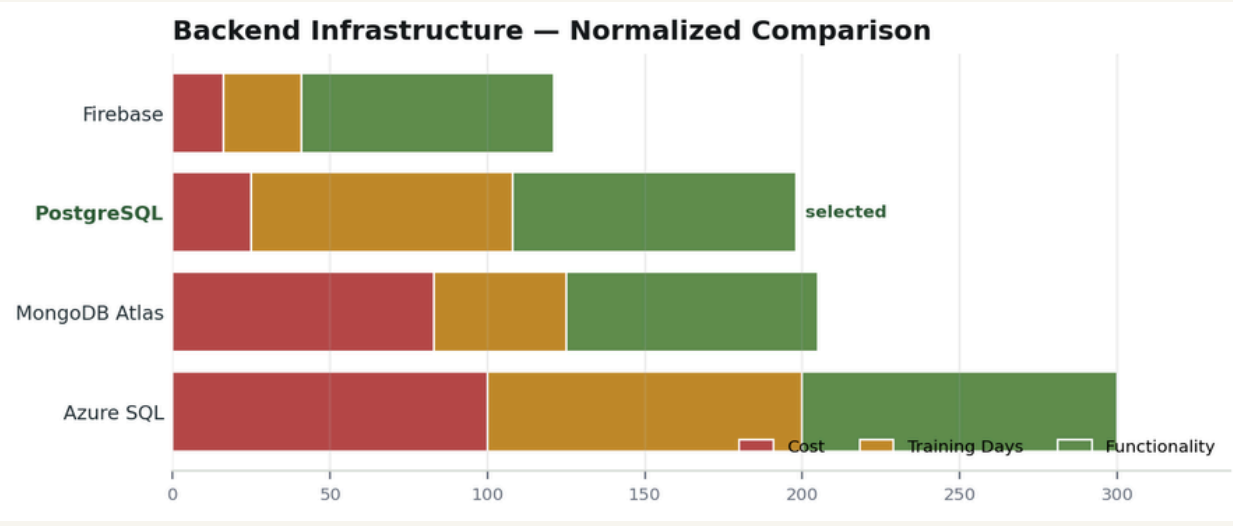
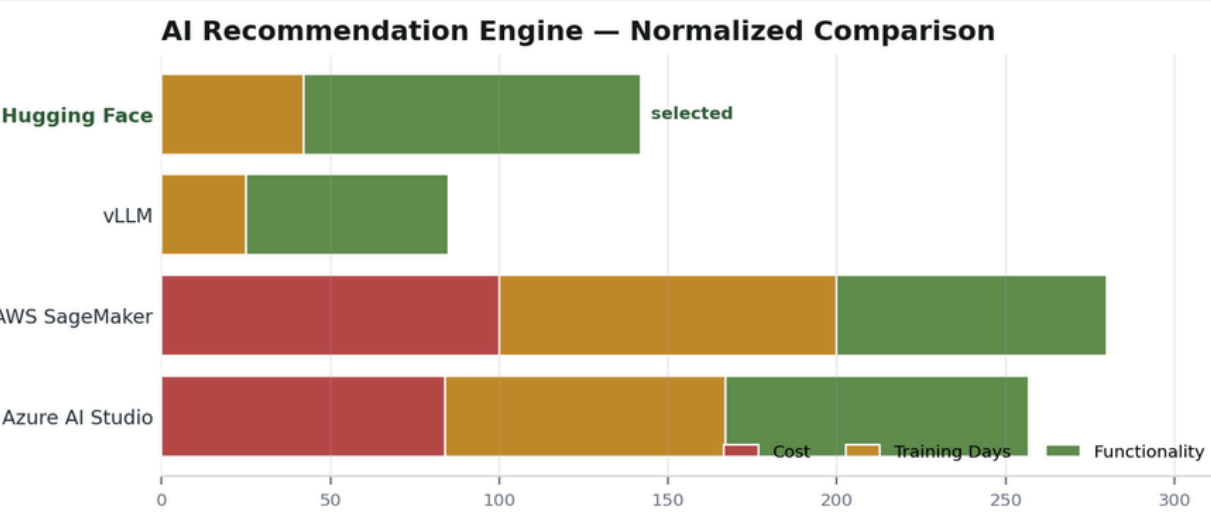
PostgreSQL

Selected for structured health, meal, workout, and user data with reliability, indexing, and complex queries.



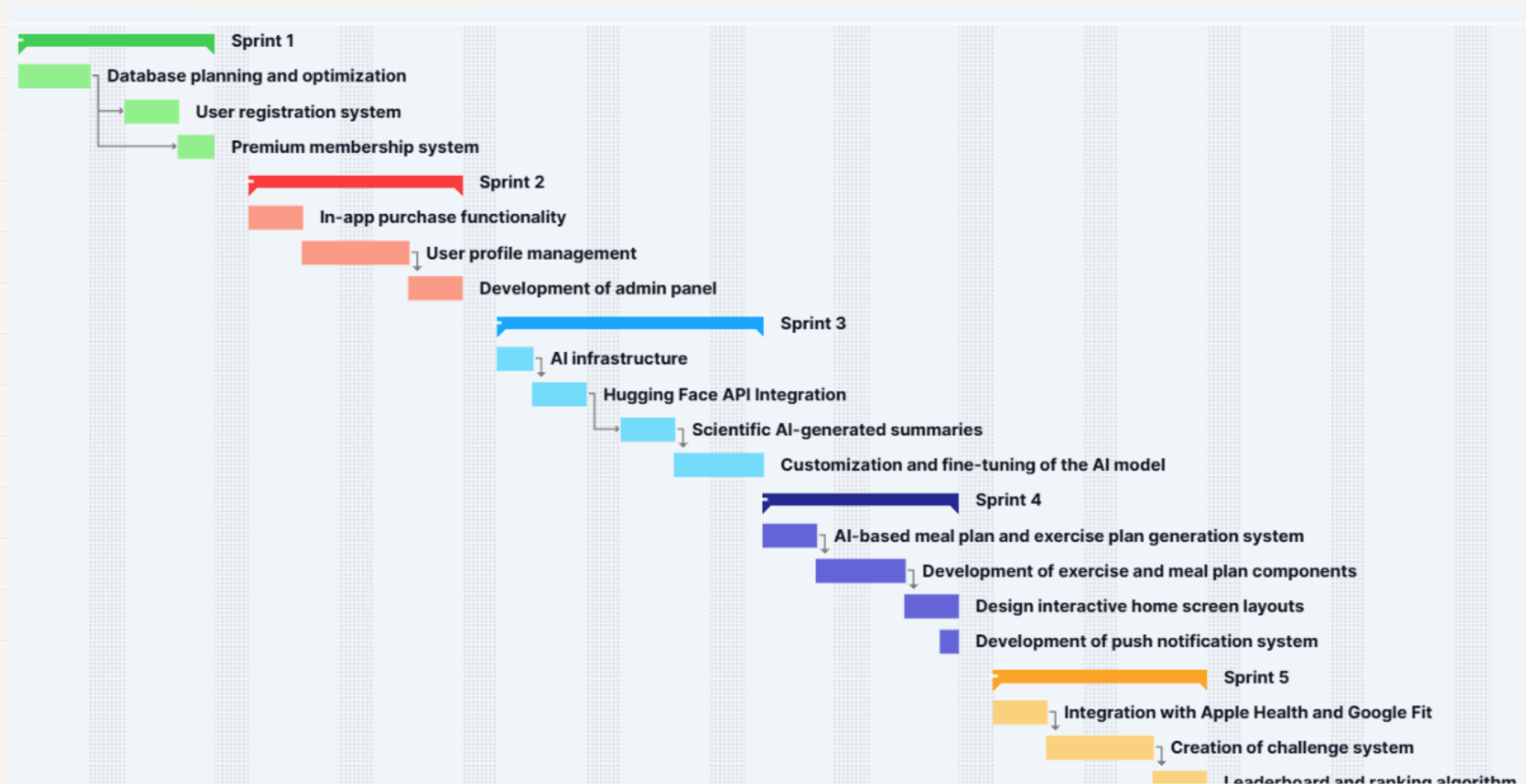
PyCharm Professional

Selected for Python AI/backend development with debugging, testing, database tools, and framework integration.



Schedule and Effort

The project is planned as a five-sprint schedule.



Stakeholders

Who is affected by project success or failure?

Users

Receive personalized coaching and motivation.

Project Owner

Benefits from growth, profit, and recognition.

Health & Fitness Experts

Support safer, more accurate guidance.

Service Providers

Provide AI APIs, hosting, and infrastructure.

Wearable Companies

Connect health data with the app.

Research Centers

Provide scientific sources for recommendations.

Gyms & Nutrition Companies

Potential partners for offers and services.

Competitors

May react by improving similar products.

Project Risks

Main risks that must be controlled.

No real behavior data

Habit prediction is difficult before enough users generate real feedback.

Low AI accuracy

Generated meal or workout plans may be ineffective or unsafe without validation.

Integration issues

AI, backend, database, and wearable data may not work smoothly together.

Compliance risk

Improper health data handling may create KVKK/GDPR problems.

Schedule delays

AI and backend tasks may take longer than expected.

API costs

AI token and service usage may exceed the planned budget.

Highest-priority risks are those with both high likelihood and high impact.

Measurements

How we track progress, quality, and product value.

Budget / Team Effort

Budget: actual cost vs.
planned budget
Team Effort: story points
completed

AI Accuracy

Correct and helpful AI
meal/workout plans divided by
total generations

Data Quality / Processing

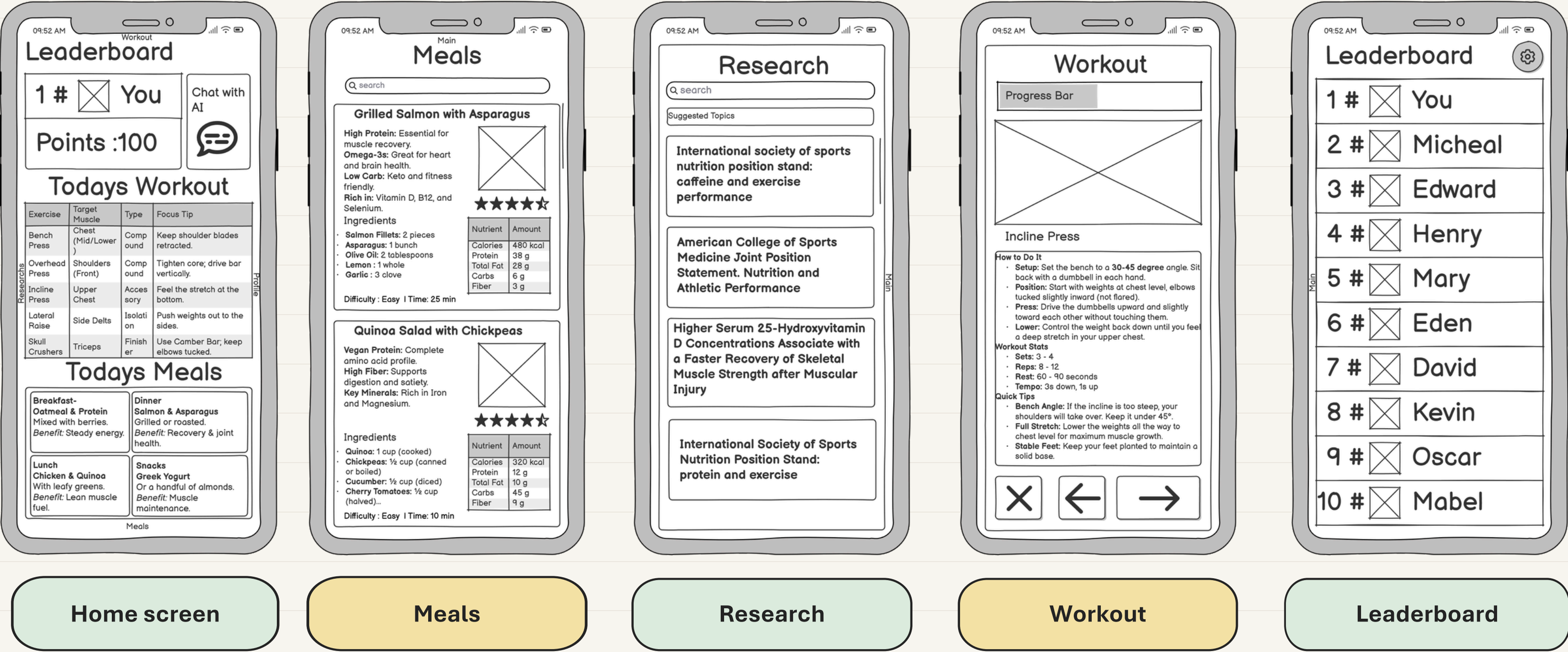
Valid sensor data per MB
Successfully processed rows /
collected rows

Defect Density

Accurate, error-free, usable
rows divided by total collected
rows

Measurements connect software project management with the actual usefulness of the product.

Graphical User Interfaces



The screens are concentrated here so they support the app story without distracting every technical slide.

Conclusion

SpotterAI aims to make personal health coaching more accessible, adaptive, and motivating.

Questions?